GEOMETRY – SHEET 6 – Surfaces and Arclength

1. Consider the hyperboloid of one sheet with equation $x^2 + y^2 = z^2 + 1$.

(i) Show that this hyperboloid can be parametrized as

$$\mathbf{r}(\theta, \lambda) = (\cos\theta, \sin\theta, 0) + \lambda\,(\sin\theta, -\cos\theta, 1) \qquad 0 \leqslant \theta < 2\pi,\ \lambda \in \mathbb{R}.$$

(ii) Determine a normal vector at $\mathbf{r}(\theta, \lambda)$ by working out $\mathbf{r}_\theta \wedge \mathbf{r}_\lambda$.

(iii) Determine a normal vector at $\mathbf{r}(\theta, \lambda)$ by working the gradient vector of $f(x, y, z) = x^2 + y^2 - z^2$ at $\mathbf{r}(\theta, \lambda)$.

(iv) What is the tangent plane to the hyperboloid at $\mathbf{r}(\theta, \lambda)$?

2. (i) What is the shortest distance between $\left(0, 1/\sqrt{2}, 1/\sqrt{2}\right)$ and $\left(1/2, 0, \sqrt{3}/2\right)$ as measured on the unit sphere $x^2 + y^2 + z^2 = 1$?

(ii) Let $\gamma(t) = (\cos\theta(t), \sin\theta(t), z(t))$, where $a \leqslant t \leqslant b$, be a parametrized curve in the cylinder $x^2 + y^2 = 1$. Show that the curve γ has arc length

$$\int_a^b \sqrt{\dot\theta^2 + \dot z^2}\,\mathrm{d}t.$$

Deduce that the map $(\theta, z) \to (\cos\theta, \sin\theta, z)$ from $\mathbb{R}^2$ to the cylinder is an isometry (when distances are measured on the cylinder).

Find the shortest distance from $(1, 0, 0)$ to $(0, 1, 1)$ when measured on the cylinder.

3. Consider the following parameterisation of the cone $x^2 + y^2 = z^2$:

$$\mathbf{r}(h, \theta) = (h\cos\theta, h\sin\theta, h).$$

Does this parameterisation represent an isometry? That is, does a curve in the h-θ plane have the same length when mapped to the cone?

4. (i) Consider the following construction of a torus: a circle of radius a is situated in the x-z plane with centre at the point $x = b$, $z = 0$, where $b > a$; and this circle is then rotated about the z axis.

Obtain the following parameterisation:

$$\mathbf{r}(\theta, \phi) = ((b + a\cos\theta)\cos\phi, (b + a\cos\theta)\sin\phi, a\sin\theta), \tag{1}$$

and describe how the values of θ and ϕ relate to position on the torus.

(ii) The planar curve $(x(s), y(s))$, $s \in (0, l)$, for which $x(s) > 0$, is rotated around the y-axis. Give a parameterisation for the resulting surface.

(iii) *Bonus:* if the curve in (ii) were rotated around, say, the line $y = mx$ for some $m > 0$, how might you go about constructing a parameterisation? (You needn't do this in detail, just outline the steps.)

5. Consider two points on a torus with equal values of the angle θ (as defined in Q4(i)) and different values of ϕ. Give an arc length parameterisation for the path connecting the points where θ remains fixed. Show that this is *not* a geodesic, except at two particular values of θ, which you should determine.

6. (i) Let $\mathbf{r}(u, v)$ be a parametrized surface with unit normal $\mathbf{n}(u, v)$. By differentiating $\mathbf{n} \cdot \mathbf{n} = 1$, show that $\mathbf{n}_u$ and $\mathbf{n}_v$ are tangent vectors to the surface. Deduce that

$$\mathbf{n}_u \wedge \mathbf{n}_v = K(u, v)\,\mathbf{r}_u \wedge \mathbf{r}_v$$

for some scalar function $K(u, v)$, known as the *Gaussian curvature*.

(ii) When $\mathbf{r}(u, v)$ is the sphere of radius R centred at $\mathbf{0}$, so that $\mathbf{r} = R\mathbf{n}$, what is $K(u, v)$?

(iii) Let $0 < a < b$. Consider the torus parameterisation (1). Determine $\mathbf{n}(\theta, \phi) = \mathbf{r}_\theta \wedge \mathbf{r}_\phi / |\mathbf{r}_\theta \wedge \mathbf{r}_\phi|$ and hence find $K(\theta, \phi)$. Where is $K(\theta, \phi)$ positive, where negative? Can you think of a surface for which K would be everywhere zero?

7. (*Optional*) Let S denote the unit sphere $x^2 + y^2 + z^2 = 1$. Every point $P = (r\cos\alpha, r\sin\alpha)$ in $\mathbb{R}^2$ can be identified with a point Q on S by drawing a line from $(r\cos\alpha, r\sin\alpha, 0)$ to the sphere's north pole $N = (0, 0, 1)$; this line intersects the sphere at two points Q and N. We define a map f from $\mathbb{R}^2$ to the sphere S by setting $f(P) = Q$.

(i) Show that

$$Q = \left(\frac{2r\cos\alpha}{1+r^2}, \frac{2r\sin\alpha}{1+r^2}, \frac{r^2-1}{1+r^2} \right).$$

(ii) What are the spherical polar co-ordinates θ, ϕ of the point Q?